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Lab 5: Developing Web Application using Django-Part I

Question:

1. Create a web form which allows the teacher to enter the student details like name, date of birth, address, contact number, email id and marks of English, Physics and Chemistry. After filling all the information and on clicking submit button, details should be added to a textarea displayed on the same page. Display the total percentage of marks obtained in a label.

Forms.py

```
from django import forms
```

```
class StudentForm(forms.Form):
    name = forms.CharField(
        max_length=100,
        widget=forms.TextInput(attrs={'class': 'form-control', 'placeholder': 'Enter student name'})
    )
    date_of_birth = forms.DateField(
        widget=forms.DateInput(attrs={'class': 'form-control', 'type': 'date'})
    )
    address = forms.CharField(
        widget=forms.Textarea(attrs={'class': 'form-control', 'rows': 3, 'placeholder': 'Enter address'})
    )
    contact_number = forms.CharField(
        max_length=15,
        widget=forms.TextInput(attrs={'class': 'form-control', 'placeholder': 'Enter contact number'})
    )
    email_id = forms.EmailField(
        widget=forms.EmailInput(attrs={'class': 'form-control', 'placeholder': 'Enter email'})
    )
```

```
english_marks = forms.IntegerField(  
    min_value=0,  
    max_value=100,  
    widget=forms.NumberInput(attrs={'class': 'form-control', 'placeholder': 'Marks (0-100)'}  
    )  
    physics_marks = forms.IntegerField(  
        min_value=0,  
        max_value=100,  
        widget=forms.NumberInput(attrs={'class': 'form-control', 'placeholder': 'Marks (0-100)'}  
        )  
        chemistry_marks = forms.IntegerField(  
            min_value=0,  
            max_value=100,  
            widget=forms.NumberInput(attrs={'class': 'form-control', 'placeholder': 'Marks (0-100)'}  
            )
```

Views.py

```
from django.shortcuts import render
from .forms import StudentForm

def student_form_view(request):
    student_details = None
    percentage = None
    if request.method == 'POST':
        form = StudentForm(request.POST)
        if form.is_valid():
            # Get cleaned data
            data = form.cleaned_data
            # Calculate percentage
            total_marks = (data['english_marks'] +
                           data['physics_marks'] +
                           data['chemistry_marks'])
            percentage = (total_marks / 300) * 100
            # Format student details for textarea
            student_details = f"""
            Name: {data['name']}
            Date of Birth: {data['date_of_birth']}
            Address: {data['address']}
            Contact Number: {data['contact_number']}
            Email ID: {data['email_id']}
            English Marks: {data['english_marks']}
            Physics Marks: {data['physics_marks']}
            Chemistry Marks: {data['chemistry_marks']}
            Total Marks: {total_marks}/300
            Percentage: {percentage:.2f}%
            """.strip()
        else:
            form = StudentForm()
            context = {
                'form': form,
                'student_details': student_details,
                'percentage': percentage
            }
```

```
return render(request, 'student_app/student_form.html', context)
```

student_form.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Student Details Form</title>
```

```
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
```

```
    <style>
```

```
        body {
```

```
            background-color: #f8f9fa;
```

```
            padding: 20px;
```

```
        }
```

```
        .container {
```

```
            max-width: 800px;
```

```
            background-color: white;
```

```
            padding: 30px;
```

```
            border-radius: 10px;
```

```
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
```

```
            margin-top: 20px;
```

```
        }
```

```
        h2 {
```

```
            color: #333;
```

```
        margin-bottom: 30px;
        text-align: center;
    }
    .percentage-label {
        font-size: 1.5rem;
        font-weight: bold;
        color: #28a745;
        padding: 10px;
        background-color: #d4edda;
        border-radius: 5px;
        text-align: center;
        margin-bottom: 20px;
    }
    .details-section {
        margin-top: 30px;
    }
</style>
</head>
<body>
    <div class="container">
        <h2> Student Details Entry Form</h2>

        <form method="post">
            {% csrf_token %}

            <div class="mb-3">
```

```
<label for="{{ form.name.id_for_label }}" class="form-label">Name:</label>
{{ form.name }}
</div>
```

```
<div class="mb-3">
    <label for="{{ form.date_of_birth.id_for_label }}" class="form-label">Date of
    Birth:</label>
    {{ form.date_of_birth }}
</div>
```

```
<div class="mb-3">
    <label for="{{ form.address.id_for_label }}" class="form-label">Address:</label>
    {{ form.address }}
</div>
```

```
<div class="mb-3">
    <label for="{{ form.contact_number.id_for_label }}" class="form-label">Contact
    Number:</label>
    {{ form.contact_number }}
</div>
```

```
<div class="mb-3">
    <label for="{{ form.email_id.id_for_label }}" class="form-label">Email ID:</label>
    {{ form.email_id }}
</div>
```

```
<div class="row">

  <div class="col-md-4 mb-3">

    <label for="{{ form.english_marks.id_for_label }}" class="form-label">English
Marks:</label>

    {{ form.english_marks }}

  </div>
```

```


  <div class="col-md-4 mb-3">

    <label for="{{ form.physics_marks.id_for_label }}" class="form-label">Physics
Marks:</label>

    {{ form.physics_marks }}

  </div>
```

```


  <div class="col-md-4 mb-3">

    <label for="{{ form.chemistry_marks.id_for_label }}" class="form-
label">Chemistry Marks:</label>

    {{ form.chemistry_marks }}

  </div>

</div>
```

```


<div class="text-center">

  <button type="submit" class="btn btn-primary btn-lg">Submit</button>

</div>

</form>
```

```
{% if percentage is not None %}

<div class="details-section">
```

```

<div class="percentage-label">

    Percentage: {{ percentage|floatformat:2 }}%

</div>

<div class="mb-3">

    <label class="form-label"><strong>Student Details:</strong></label>

    <textarea class="form-control" rows="12" readonly>{{ student_details }}</textarea>

</div>

</div>

{% endif %}

</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></sc
ript>

</body>

</html>

```

Urls.py

```

from django.urls import path

from . import views

urlpatterns = [

    path("", views.student_form_view, name='student_form'),

]

```


OUTPUT:



Student Details Entry Form

Name:

Date of Birth:



Address:

Contact Number:

Email ID:

English Marks:

Physics Marks:

Chemistry Marks:

Submit

2> Create a web form with employee ids in dropdown list. Use a textbox for accepting date of joining. Add a button named “Am I Eligible for Promotion”. On clicking the button, if he has more than 5 years of experience, then display “YES” else “NO” in the label control.

Forms.py

```
from django import forms
```

```
from datetime import date
```

```
# Sample employee data (you can replace this with database data)
```

```
EMPLOYEE_CHOICES = [
```

```
    ('', 'Select Employee ID'),
```

```
    ('EMP001', 'EMP001'),
```

```
    ('EMP002', 'EMP002'),
```

```
    ('EMP003', 'EMP003'),
```

```
    ('EMP004', 'EMP004'),
```

```
    ('EMP005', 'EMP005'),
```

```
    ('EMP006', 'EMP006'),
```

```
    ('EMP007', 'EMP007'),
```

```
    ('EMP008', 'EMP008'),
```

```
    ('EMP009', 'EMP009'),
```

```
    ('EMP010', 'EMP010'),
```

```
]
```

```
class PromotionEligibilityForm(forms.Form):
```

```
    employee_id = forms.ChoiceField(
```

```
        choices=EMPLOYEE_CHOICES,
```

```
        widget=forms.Select(attrs={'class': 'form-control'})
```

```
)  
date_of_joining = forms.DateField(  
    widget=forms.DateInput(attrs={'class': 'form-control', 'type': 'date'})  
)
```

```
def calculate_experience(self):  
    """Calculate years of experience"""  
    if self.is_valid():  
        doj = self.cleaned_data['date_of_joining']  
        today = date.today()  
        years = today.year - doj.year  
  
        # Adjust if birthday hasn't occurred this year  
        if (today.month, today.day) < (doj.month, doj.day):  
            years -= 1  
  
        return years  
    return 0
```

Views.py

```
from django.shortcuts import render

from .forms import PromotionEligibilityForm

from datetime import date


def promotion_eligibility_view(request):

    result = None

    years_of_experience = None

    employee_id = None


    if request.method == 'POST':

        form = PromotionEligibilityForm(request.POST)

        if form.is_valid():

            # Get cleaned data

            employee_id = form.cleaned_data['employee_id']


            # Calculate years of experience

            years_of_experience = form.calculate_experience()


            # Check eligibility (more than 5 years)

            if years_of_experience > 5:

                result = "YES"

            else:

                result = "NO"

        else:

            form = PromotionEligibilityForm()
```

```
context = {  
    'form': form,  
    'result': result,  
    'years_of_experience': years_of_experience,  
    'employee_id': employee_id  
}
```

```
return render(request, 'student_app/promotion_form.html', context)
```

promotion_form.html

```
<!DOCTYPE html>  
  
<html lang="en">  
  
<head>  
  
    <meta charset="UTF-8">  
  
    <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  
    <title>Promotion Eligibility Checker</title>  
  
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"  
rel="stylesheet">  
  
<style>  
  
    body {  
  
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  
  
        min-height: 100vh;  
  
        padding: 20px;  
  
        font-family: 'Arial', sans-serif;  
  
    }
```

```
.container {  
    max-width: 600px;  
    background-color: white;  
    padding: 40px;  
    border-radius: 15px;  
    box-shadow: 0 10px 30px rgba(0,0,0,0.3);  
    margin-top: 50px;  
}
```

```
h2 {  
    color: #333;  
    margin-bottom: 30px;  
    text-align: center;  
    font-weight: bold;  
}
```

```
.result-section {  
    margin-top: 30px;  
    text-align: center;  
}
```

```
.result-label {  
    font-size: 3rem;  
    font-weight: bold;  
    padding: 30px;  
    border-radius: 10px;  
    margin: 20px 0;  
    animation: fadeIn 0.5s;  
}
```

```
.result-yes {  
    background-color: #d4edda;  
    color: #155724;  
    border: 3px solid #28a745;  
}  
  
.result-no {  
    background-color: #f8d7da;  
    color: #721c24;  
    border: 3px solid #dc3545;  
}  
  
.experience-info {  
    font-size: 1.2rem;  
    color: #666;  
    margin-bottom: 15px;  
}  
  
.btn-check {  
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);  
    border: none;  
    padding: 12px 40px;  
    font-size: 1.1rem;  
    font-weight: bold;  
}  
  
.btn-check:hover {  
    background: linear-gradient(135deg, #764ba2 0%, #667eea 100%);  
    transform: translateY(-2px);  
    box-shadow: 0 5px 15px rgba(0,0,0,0.2);
```



```
}  
  
@keyframes fadeIn {  
  from { opacity: 0; transform: scale(0.8); }  
  to { opacity: 1; transform: scale(1); }  
}
```

```
.form-label {  
  font-weight: 600;  
  color: #333;  
  margin-bottom: 8px;  
}
```

```
.icon {  
  font-size: 2rem;  
  margin-right: 10px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Promotion Eligibility Checker</h2>
```

```
<form method="post">
```

```
{% csrf_token %}
```

```
<div class="mb-4">
```

```
<label for="{{ form.employee_id.id_for_label }}" class="form-label">
```

 Employee ID:

```

</label>

{{ form.employee_id }}

{% if form.employee_id.errors %}

    <div class="text-danger mt-1">{{ form.employee_id.errors }}</div>

{% endif %}

</div>


<div class="mb-4">

    <label for="{{ form.date_of_joining.id_for_label }}" class="form-label">

        Date of Joining:

    </label>

    {{ form.date_of_joining }}

    {% if form.date_of_joining.errors %}

        <div class="text-danger mt-1">{{ form.date_of_joining.errors }}</div>

    {% endif %}

</div>


<div class="text-center">

    <button type="submit" class="btn btn-primary btn-check">

        Am I Eligible for Promotion?

    </button>

</div>

</form>


{% if result is not None %}

<div class="result-section">

```

```
<div class="experience-info">

    <strong>Employee ID:</strong> {{ employee_id }}<br>

    <strong>Years of Experience:</strong> {{ years_of_experience }}
year{{ years_of_experience|pluralize }}

</div>


<div class="result-label {% if result == 'YES' %}result-yes{% else %}result-no{%
endif %}">

    {% if result == "YES" %}

        {{ result }}

    {% else %}

        {{ result }}

    {% endif %}

</div>


<p class="text-muted">

    {% if result == "YES" %}

        Congratulations! You are eligible for promotion. 🎉

    {% else %}

        You need more than 5 years of experience for promotion eligibility.

    {% endif %}

</p>

</div>

{% endif %}

</div>
```

```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></sc
ript>

</body>

</html>
```

Urls.py

```
from django.urls import path

from . import views
```

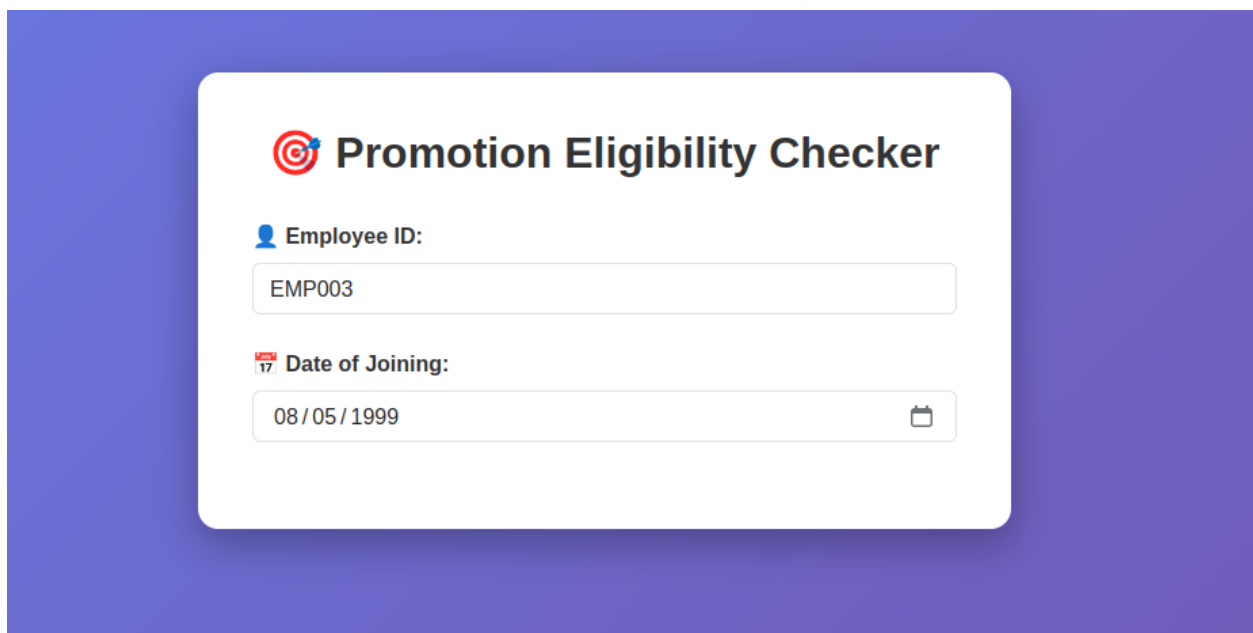
```
urlpatterns = [

    path("", views.student_form_view, name='student_form'),

    path('promotion/', views.promotion_eligibility_view, name='promotion_eligibility'),

]
```

OUTPUT:



The screenshot shows a web application titled "Promotion Eligibility Checker" with a red target icon. It features two input fields: "Employee ID:" with the value "EMP003" and "Date of Joining:" with the value "08/05/1999". The date field includes a calendar icon. The interface is set against a solid purple background.